



CONFIGURATION MANAGEMENT PLAN

LITPATH AI: SMART PATHFINDER FOR THESES AND DISSERTATIONS

ASIA PACIFIC COLLEGE
3 HUMABON, MAGALLANES
MAKATI CITY, METRO MANILA 1232

APRIL 2026





TABLE OF CONTENTS

Introduction.....	3
Roles and Responsibilities	4
Project Sponsor.....	4
Project Manager	5
Configuration Manager (DevOps Engineer)	5
Developers (Frontend and Backend)	5
QA Tester	5
Configuration Control	6
Configuration Management Database (CMDB).....	7
CMDB Tools	8
CMDB Access Control	8
Configuration Status Accounting.....	8
1) Configuration Change Requests (CCRs).....	9
2) Version Control Status.....	9
3) Build and Release Reporting	9
4) Audit Status	10
Configuration Audits.....	10





Introduction

The purpose of the Configuration Management Plan is to describe how configuration management (CM) will be conducted throughout the project lifecycle. This includes documenting how CM is managed, roles and responsibilities, how configuration item (CI) changes are made, and communicating all aspects of CM to project stakeholders. Without a documented configuration management plan it is likely that CIs may be missed, incomplete, or unnecessary work is done because of a lack of version and document control. While a configuration management plan is important for all projects, this is especially so for software and other information technology (IT) projects.

The purpose of this Configuration Management Plan (CMP) is to describe how configuration management (CM) will be conducted throughout the lifecycle of the **LitPath AI** project. This plan documents how configuration management activities are managed, defines roles and responsibilities, establishes how configuration item (CI) changes are identified and controlled, and describes how all configuration-related information is communicated to project stakeholders. Without a formally documented configuration management plan, configuration items may be overlooked, improperly versioned, or inconsistently maintained, resulting in rework, deployment errors, or loss of system integrity. While configuration management is important for all projects, it is especially critical for **software and information technology projects** such as LitPath AI.

The LitPath AI project utilizes a modern, hybrid technology environment that combines **on-premise infrastructure hosted on Proxmox VE** with secure external access provided through **Cloudflare Tunnel**. The system integrates web-based application components, artificial intelligence services, relational and vector databases, and institutional research data owned by DOST-STII. This environment enables remote access, centralized administration, AI-powered search and analytics, and improved monitoring of system usage and performance. As a result, DOST-STII benefits from improved accessibility to academic resources, enhanced research efficiency, and reduced operational effort compared to traditional keyword-based catalog systems.

In order to effectively manage the LitPath AI project, a coordinated and well-defined Configuration Management Plan is required. This CMP establishes the configuration management framework that the project team will use to identify, track, implement, audit, and communicate configuration items and configuration changes throughout the project lifecycle. By enforcing standardized configuration control, version management, and audit practices, this plan ensures system stability, supports secure and reliable deployments, and enables informed decision-making by the Project Manager and Project Sponsor.



Roles and Responsibilities

Roles and responsibilities is an important part of any plan. In order to communicate a clear understanding of expectations, these roles and responsibilities must be clearly defined. Any work that will be performed as part of the plan must be assigned to someone, and this section allows us to illustrate who owns these tasks and to communicate them to all project stakeholders.

Clear roles and responsibilities are essential to ensure effective configuration management. All CM-related activities for LitPath AI are assigned to specific project roles as defined below.

Configuration Control Board (CCB)

The Configuration Control Board (CCB) is responsible for reviewing and approving all significant configuration changes.

CCB Members:

- Project Sponsor (Ms. Khasian Eunice Romulo)
- Project Manager (Ms. Tracie Tomon)
- DevOps Engineers (Charijoy Cempron, Joshua Sta.Ana,)
- Developers (Ms. Tracie Tomon, Ms. Jenine Dulay, Charijoy Cempron, Joshua Sta.Ana)
- QA Tester (Jenine Dulay, Joshua Sta.Ana)

CCB Responsibilities:

- Review and approve or reject Configuration Change Requests (CCRs)
- Evaluate the impact of changes on scope, schedule, cost, quality, and risk
- Ensure approved changes are recorded in the CMDB

Project Sponsor

The Project Sponsor is responsible for:

- Providing final approval for configuration changes that affect project scope, schedule, or cost
- Participating in CCB decision-making as required



Project Manager

The Project Manager is responsible for:

- Overall accountability for configuration management activities
- Identifying configuration items
- Communicating CM activities and decisions to stakeholders
- Re-baselining project artifacts when approved changes are implemented
- Participating in CCB reviews

Configuration Manager (DevOps Engineer)

The Configuration Manager is responsible for:

- Managing and maintaining the Configuration Management Database (CMDB)
- Defining and enforcing version control standards
- Managing infrastructure, deployment, and environment configurations
- Conducting configuration audits prior to releases

Developers (Frontend and Backend)

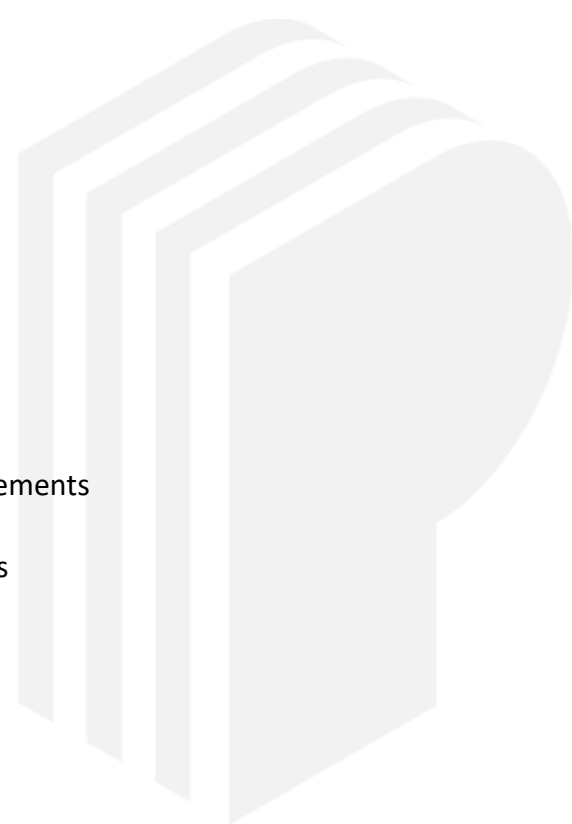
Developers are responsible for:

- Implementing approved configuration changes
- Ensuring changes comply with version control standards
- Documenting changes and their impacts
- Supporting testing and rollback procedures

QA Tester

The QA Tester is responsible for:

- Verifying that configuration changes meet approved requirements
- Performing functional and regression testing after changes
- Supporting configuration audits through validation activities





Configuration Control

Configuration Control is the process of systematically controlling and managing all steps of configuration throughout the project lifecycle. In order to effectively handle project CM it is important to use a process which ensures only necessary configuration changes are made. Additionally, like any change management efforts, configuration change decisions must be made with the understanding of the impact of the change.

Configuration Control is the process of systematically controlling and managing all configuration activities throughout the lifecycle of the LitPath AI project. To effectively manage configuration items (CIs), it is essential that a structured and standardized process is followed to ensure that only necessary and authorized configuration changes are implemented. Like all change management activities, configuration control decisions must be made with a clear understanding of the potential impact on project scope, schedule, cost, quality, and risk.

The LitPath AI project will utilize a standardized configuration control process for the duration of the project lifecycle to ensure that all configuration items are handled consistently, reviewed thoroughly, and communicated appropriately to all stakeholders. This approach supports system stability, traceability, and accountability, particularly given the project's use of AI components, hybrid infrastructure (Proxmox VE and Cloudflare Tunnel), and institutional data owned by DOST-STII.

As configuration items are identified by the project team, the Configuration Manager (DevOps Engineer) will assign a unique CI identifier and record the CI in the Configuration Management Database (CMDB) with an initial status of *Initiated*. Once recorded, the CI will be assigned to the appropriate responsible role or focus group (Frontend Developer, Backend Developer, QA Tester, or DevOps Engineer) based on the nature of the configuration item.

Authorized team members will be granted access to the CI through the CMDB to perform approved updates or modifications. Each change or edit must be clearly documented in the CMDB log, including a description of the modification, rationale for the change, date, and responsible individual. This ensures full traceability of all configuration changes throughout the project lifecycle.

It is imperative that all software-related configuration changes undergo appropriate testing to validate that the implemented changes function as intended and do not introduce defects or regressions. The responsible developer or focus group is required to conduct unit, integration, or functional testing as applicable. The Configuration Manager, in coordination with the QA Tester, is responsible for confirming that testing has been completed, results are documented,



and changes are properly recorded in the CMDB. The Configuration Manager is also responsible for assigning updated version numbers and revising the CI status to reflect its current state.

Configuration items often maintain dependencies or relationships with other CIs within the LitPath AI system, such as relationships between frontend components, backend services, databases, and infrastructure configurations. The Configuration Manager, Project Manager, and responsible developers will collaborate to identify and document these relationships. All CI dependencies and co-dependencies will be illustrated and maintained within the CMDB to ensure a comprehensive understanding of system interactions and potential downstream impacts of changes.

Any configuration changes identified by project team members or stakeholders must be formally documented using a Configuration Change Request (CCR) and submitted to the Configuration Control Board (CCB) for review. The CCB will analyze each CCR and approve or reject the request based on its assessed impact on scope, schedule, cost, quality, and risk. If a configuration change is approved, project baselines and requirements will be updated as necessary, and the Project Manager will ensure that all approved changes are communicated to relevant stakeholders. CCRs that are denied may be revised and resubmitted with additional justification or supporting information for reconsideration by the CCB. No configuration change may be deployed to the production environment without formal CCB approval and proper documentation within the CMDB.

Configuration Management Database (CMDB)

A Configuration Management Database (CMDB) is where the organizations configuration information is stored. CMDB is a term which originates from Information Technology Infrastructure Library (ITIL) which provides a framework for best practices in IT services management. The CMDB contains not only the configuration information for assets but also information about the assets such as physical location, ownership, and its relationship to other configurable items (CIs).

A key component to configuration management is having a well defined and followed process for both document and data management.

The Configuration Management Database (CMDB) serves as the centralized repository for all configuration information related to LitPath AI.

The CMDB contains:



- Configuration item identifiers
- Version numbers and baselines
- Change history and audit trails
- Relationships between configuration items

CMDB Tools

The LitPath AI CMDB is implemented using:

- Git repositories for source code and documentation
- Secure storage for environment variables and secrets
- Deployment and infrastructure configuration records

CMDB Access Control

Access to the CMDB is controlled as follows:

- Read/Write Access: Project Manager, Configuration Manager, Developers
- Read-Only Access: Project Sponsor and DOST-STII stakeholders

The CMDB ensures that all team members are working with the most current approved configuration while maintaining historical records for traceability.

Configuration Status Accounting

Accounting for the status of the configuration involves the collection, processing, and reporting of the configuration data for all CIs at any given time. This also includes management stored configuration information held in the Configuration Management Database (CMDB). This may include approved configuration documents, software, data, and their current version numbers; build reports; status of any submitted changes; or any discrepancies and status identified through configuration audits.

Configuration status accounting is the process of collecting, processing, and reporting configuration information for all configuration items (CIs) at any given point in time. This activity includes the management and reporting of configuration data stored within the Configuration Management Database (CMDB), such as approved configuration documents, software components, infrastructure configurations, data assets, and their associated version numbers. Configuration status accounting also provides visibility into build reports, the status



of submitted Configuration Change Requests (CCRs), and any discrepancies or issues identified through configuration audits.

For the LitPath AI project, configuration status accounting is critical to ensure transparency, traceability, and informed decision-making across the project lifecycle. It is essential that the Project Sponsor (DOST-STII) and the Project Manager have the ability to review the current configuration status at any time. To support this requirement, the Project Manager will include configuration status information as part of regular project reporting.

The Project Manager will submit weekly project status reports, which will include a dedicated configuration status section. This section will provide a consolidated view of the current state of configuration items and changes. Configuration status reporting will include, but is not limited to, the following categories:

1) Configuration Change Requests (CCRs)

- Aging: The length of time each CCR has remained open, from submission to resolution
- Distribution: The number of CCRs submitted, categorized by role or functional area (Frontend, Backend, DevOps, QA)
- Trending: Identification of areas where configuration changes are most frequently occurring, helping highlight risk or instability within specific components

2) Version Control Status

Configuration version status will be tracked and reported for all major CI categories, including:

- Software: Frontend applications, backend services, APIs, and AI components
- Infrastructure: Proxmox VE configurations, Cloudflare Tunnel settings, and deployment scripts
- Data: Database schemas, migration scripts, and AI embedding datasets
- Documentation: Project plans, technical documentation, and user/admin guides

3) Build and Release Reporting

Build reporting provides visibility into the composition and readiness of system releases and includes:

- Files: Confirmation that all required source code and configuration files are included in the build



- CI Relationships: Verification that dependencies and relationships between configuration items are correctly represented and compatible
- Incorporated Changes: A summary of approved CCRs and enhancements included in each build or release

https://asiapacificcollege.sharepoint.com/:w:/r/sites/HecTech2/_layouts/15/Doc.aspx?sourcedoc=%7BAEF50DB7-5F83-4961-B8D1-3C2E45A612EC%7D&file=04%20Project%20Monitoring%20and%20Controlling%2002%20Project-Status-Report.docx&action=default&mobileredirect=true

4) Audit Status

Configuration status accounting also tracks the results and status of configuration audits, including:

- Physical Configuration Status: Verification that deployed components match documented and approved configurations
- Functional Configuration Status: Verification that implemented features meet approved requirements and specifications

Prior to any new software release, the Configuration Manager (DevOps Engineer) will work closely with the Project Manager and responsible developers to ensure that all configuration items are updated to their latest approved release versions, all CCRs have been properly incorporated, and the CMDB accurately reflects the current system baseline.

CONFIGURATION AUDITS

Audits are an important part of project and configuration management. The purpose of an audit is to ensure that established processes are being followed as intended and to provide an opportunity to correct any deviations from these processes. Many people hold a negative view of audits; however, when used appropriately, audits are an effective management and quality assurance tool.

Configuration audits are an essential component of both project management and configuration management for the LitPath AI project. The primary purpose of a configuration audit is to ensure that established configuration management processes and procedures are being followed as intended and to provide a structured opportunity to identify, document, and correct any deviations from these processes. While audits are sometimes viewed negatively, when applied appropriately, they serve as an effective quality assurance and management control mechanism that helps maintain system integrity, reliability, and compliance.



Configuration audits will be an ongoing activity throughout the LitPath AI project lifecycle. The objective of each configuration audit is to verify that all project team members are adhering to approved configuration management standards, version control practices, and documentation requirements. Formal configuration audits will be conducted prior to any major software release, as well as at the discretion of the Project Manager or Project Sponsor should they determine that an audit is necessary due to risk, scope changes, or quality concerns.

All configuration audits for the LitPath AI project will be performed by the Configuration Manager (DevOps Engineers). Throughout the project, the Configuration Manager works closely with the Frontend Developers, Backend Developers, QA Testers, and Project Manager to ensure that configuration management procedures are consistently applied. As part of each configuration audit, the Configuration Manager will perform the following activities:

- Establish a dedicated audit environment within the Configuration Management Database (CMDB)
- Copy the latest approved versions of all software components, databases, infrastructure configurations, and project documentation into the audit environment
- Verify that all configuration items are correctly versioned and that version control standards have been properly applied
- Analyze historical versions, commit logs, and timestamps of software, data, and documentation to confirm that all changes were formally approved, recorded, and traceable
- Deploy and test the latest software versions to ensure that approved requirements and functional specifications are being met
- Verify that all required artifacts (source code, configuration files, documentation, test results) are present, complete, and current within the CMDB
- Confirm that all approved Configuration Change Requests (CCRs) have been fully implemented and accurately recorded in the CMDB

Upon completion of each configuration audit, the Configuration Manager will compile a detailed audit findings report. For each identified discrepancy, the Configuration Manager will work collaboratively with the Project Manager and relevant team members to define the



required corrective action(s), assign responsibility for each action, and establish a resolution timeline.

Once all findings have been reviewed, the Configuration Manager will document all discrepancies, corrective actions, and resolutions in a formal audit report. This report will be submitted to the Project Manager and Project Sponsor for review and approval. Audit results may be used to improve configuration management processes, reduce future risks, and support informed decision-making for subsequent project phases.





SPONSOR ACCEPTANCE

Approved by the Project Sponsor:

Date: April 30, 2026

Khasian Eunice Romulo
Science Research Specialist II / Unit Head
(Digital Information and Processing Delivery Unit)

